FINITELY PRESENTED MODULES DESCEND THROUGH LIMITS OF SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2238-2262, following the statement at 2229-2236.

Editorial note. No omissions; the formal conclusion is by the displayed descent-data identification.

Allowed standard prerequisites (editorial).

- Presentation of the limit space. lemma-directed-inverse-system-has-limit (same chapter, card L26291, proof cited at ~2243-2246). X = U/R with U = lim U_i and R = lim R_i; also the fibre-product identity R x_{s,U,t} R = colim R_i x_{s_i,U_i,t_i} R_i.
- QCoh of limits of schemes. Limits, Lemma limits-lemma-descend-modules-finite-presentation (cross-chapter, cited at ~2250-2256). The three category colimit identifications for U, R and the double fibre product.
- QCoh as descent data. Properties of Spaces, Proposition spaces-properties-proposition-quasi-coherent (cross-chapter, cited at ~2257-2260). QCoh(O_X) = QCoh(U, R, s, t, c) and likewise for X_i.

Source reference. Original lemma 07SK: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2148–2148. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07SF: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 878–878. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07V7: focused-seeds/geometry/S9-spaces-limits-descend-modules-finite-presentation/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03M3: focused-seeds/geometry/S9-spaces-limits-descend-modules-finite-presentation/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *With notation and assumptions as in Lemma 07SK. The category of $\mathcal{O}_X$-modules of finite presentation is the colimit over I of the categories $\mathcal{O}_{X_i}$-modules of finite presentation.*

Proof. Choose $0 \in I$. Choose an affine scheme U_0 and a surjective étale morphism $U_0 \to X_0$. Set $U_i = X_i \times_{X_0} U_0$. Set $R_0 = U_0 \times_{X_0} U_0$ and $R_i = R_0 \times_{X_0} X_i$. Denote $s_i, t_i : R_i \to U_i$ and $s, t : R \to U$ the two projections. In the proof of Lemma 07SF we have seen that there exists a presentation $X = U/R$ with $U = \lim U_i$ and $R = \lim R_i$. Note that U_i and U are affine and that R_i and R are quasi-compact and separated (as X_i is quasi-separated). Moreover, it is also true that $R \times_{s,U,t} R = \operatorname{colim} R_i \times_{s_i,U_i,t_i} R_i$. Thus we know that $QCoh(\mathcal{O}_U) = \operatorname{colim} QCoh(\mathcal{O}_{U_i})$, $QCoh(\mathcal{O}_R) = \operatorname{colim} QCoh(\mathcal{O}_{R_i})$, and $QCoh(\mathcal{O}_{R \times_{s,U,t} R}) = \operatorname{colim} QCoh(\mathcal{O}_{R_i \times_{s_i,U_i,t_i} R_i})$ by Limits, Lemma 07V7. We have $QCoh(\mathcal{O}_X) = QCoh(U, R, s, t, c)$ and $QCoh(\mathcal{O}_{X_i}) = QCoh(U_i, R_i, s_i, t_i, c_i)$, see Properties of Spaces, Proposition 03M3. Thus the result follows formally. □